CAPSTONE 31 — AI Verification & Ethics Report

Hallucination Audit, Outdated Syntax Remediation & DevOps Ethics

1. AI Output Verification & Hallucination Audit (Topic 10)

During multi-tool prompting across ChatGPT, Google Gemini, Claude, and DeepSeek, all AI-generated Git commands, shell scripts, and GitHub Actions YAML configurations were audited against the Pro Git Book and GitHub Docs Secure Use Reference. Five critical defects were caught and corrected before deployment:

Audit ID	Unverified AI Output	Root Cause & Risk Analysis	Verified Fix Implemented
V-01 Deprecated Action Tags	`uses: actions/checkout@v2` `uses: actions/setup-python@v2`	Trained on legacy repositories; Node 12/16 runtimes are deprecated on modern GitHub runners and fail workflow runs.	Updated to `actions/checkout@v4` and `actions/setup-python@v5` with native `cache: "pip"` (GitHub Docs).
V-02 Disabled Workflow Command	`run: echo "::set-output name=res::pass"`	`::set-output` was disabled by GitHub due to stdout injection vulnerabilities.	Replaced with environment file syntax: `echo "res=pass" >> "$GITHUB_OUTPUT"`.
V-03 Unsafe History Rewrite	`git push --force origin main`	Overwrites teammates' merged commits on the shared default branch and destroys audit history.	Blocked force-pushes on `main` via branch protection; documented `git push --force-with-lease` for feature branches only.
V-04 Script Injection in YAML	`run: echo "${{ github.event.pull_request.title }}"`	Direct shell interpolation of untrusted PR titles enables arbitrary remote code execution on the runner.	Bound untrusted context to an intermediate environment variable (`env: PR_TITLE: ...`) and quoted `"$PR_TITLE"`.

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V-05 Hallucinated Git Flag	`git branch --create-remote origin/feat`	Fabricated CLI flag combining branch creation and upstream tracking.	Verified via `git switch -c feat/name` and `git push -u origin feat/name` (Git Cheat Sheet).

2. Responsible AI & DevOps Ethics Guide (Topic 12)

- **Never Commit Secrets**: Never store API keys (OpenAI `sk-`, Google `AIza`, AWS `AKIA`, GitHub `ghp_`), database passwords, or `.env` files in Git commits. Committed secrets remain permanently in `.git/objects` even if deleted in a later commit. Always exclude `.env` in `.gitignore`, enforce pre-commit/commit-msg secret scanning (implemented in `scripts/commit-msg`), and inject CI secrets via GitHub Encrypted Secrets (`${{ secrets.KEY_NAME }}`).
- **Verify AI-Generated YAML & Enforce Least Privilege**: Always audit AI-generated workflows for deprecated runners, unpinned third-party actions, and overly permissive tokens. Explicitly set `permissions: contents: read` at the top of `.github/workflows/ci.yml` so the `GITHUB_TOKEN` cannot modify repository contents.
- **Credit Templates, Documentation & AI Tools**: Maintain academic and open-source integrity by citing workflow templates (GitHub Python CI Template), reference texts (Pro Git Book, CC BY-NC-SA 3.0), and AI prompts in the project documentation and PR checklist.